

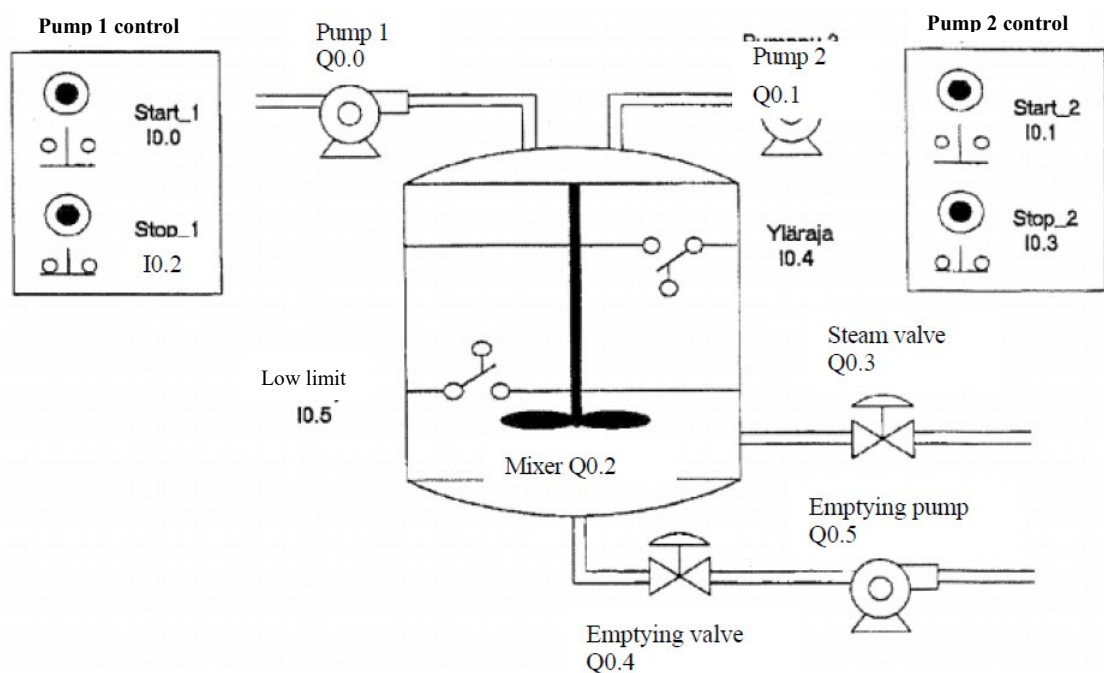
Basics of automation technology

TIA Portal exercise 4 Mixing tank

Develop and test a program that implements the operation of the attached application example as shown. Use TIA Portal software and block diagram programming (FBD) to build and simulate the application. Draw up the program in such a way that the process control is in its own function and the counters in a separate function.

Action Description:

Two different liquids are supplied to the mixing tank. Both liquids have their own pumps. The pumps are started from their own start push-buttons (closing contact, spring return). The stopping of the pumps is carried out either from the stop push-button (opening contact, spring return) or from the upper limit of the tank. The mixer in the tank turns on and the steam valve (heating) opens from the upper limit of the tank, and both are on for 100 seconds. After the mixer stops, the exhaust valve opens and the drain pump turns on. The drain pump stops and the exhaust valve closes when the lower limit of the tank is reached. Filling and emptying times are calculated using two counters. The total number of fillings should be reflected as an integer in the word MW10. Drains, respectively, in the word MW20.



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Use the following symbols and addresses:

Start_1	I0.0
Start_2	I0.1
Stop_1	I0.2
Stop_2	I0.3
High_limit	I0.4
Low_limit	I0.5
Pump1	Q0.0
Pump2	Q0.1
Mixer	Q0.2
Steam_valve	Q0.3
Eptying_valve	Q0.4
Eptying_pump	Q0.5

TESTING AND RETURNS:

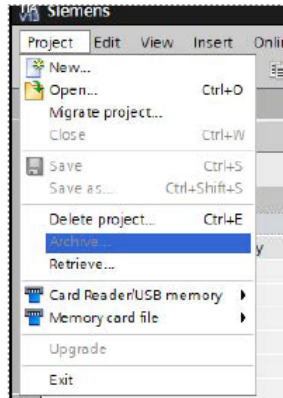
Test the operation of the program on the simulator by monitoring it. Develop a simple testing/simulation plan for testing. This may be eg. A list of numbered lines in which you have briefly recorded in a sentence or two the order in which you control the simulator's inputs during testing.

Return a **PDF** report of the work to the Moodle return box, which presents the program/solutions made in the task in the form of screenshots and contains a brief description of the challenges and successes of the task. Attach to the report the testing/simulation plan you have made, which shows how and in what order you have simulated the program.

Also, restore the project as a compressed file. To compress a project, select Project --> Archive and select a save folder.

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Networks made must be properly named/commented on and all signals/variables must have the correct symbol name. Pay attention to the commenting of the program code, ideally every Network is commented on or at least titled in a sensible way.

Also attach to the "report" screenshots of the simulation step by step so that the image shows the simulator signals and the monitoring view from the FBD program corresponding to the situation in question (can be in the same image if they fit)